

Notes on Placement for B.S.E. Students, 2025

The discussion below presumes familiarity with the [AP Table for the Class of 2029](#), which lists the course equivalences and placement recommendations of the academic departments that accept standardized test scores for Princeton requirement fulfillment and placement. You should also review the Math department's excellent [website on math placement](#). This document has a lot of information, so please take time to read it carefully and ask for clarification if something seems unclear by e-mailing BSEprogram@princeton.edu. If you find conflicting information elsewhere, please let Dean Swanson know.

The principal subject areas of interest for incoming B.S.E. students are Math, Physics, and Chemistry. Satisfaction of general B.S.E. math and science requirements through standardized test scores and Princeton placement test scores does not reduce the number of courses for graduation. Instead, course requirements that are satisfied through eligible test scores may be substituted in a student's program of study with other courses. These could be courses in the social sciences and humanities to permit wider curricular exploration or sophomore-level engineering-science courses to accelerate a departmental program of study. Appropriate substitutions should be discussed with your faculty adviser.

If you do not place out of any requirements, please don't worry! The B.S.E. program of study can be started in Physics 103, Math 103, and Chemistry 201/207 and completed on schedule. Students with no previous coursework in calculus or physics should discuss a plan with Dean Swanson that may involve summer study before the sophomore year.

Departmental Placement Tests in Physics and Chemistry in 2025

The following information explains the policies effective for the fall of 2025 for the placement tests offered by Physics and Chemistry. Please direct all questions about these tests to those departments. Placement exams do not go on any permanent record, so there's no harm in trying them.

Physics: Students seeking place out PHY103-104 (the BSE physics requirement) must take the departmental requirement fulfillment (placement) test this summer. **You cannot place out of Physics 103-104 without taking the departmental requirement fulfillment test**, no matter how high a score you got on the AP/IB/A-levels or any other preparation (pre-2020 webpages might say otherwise). There is a separate exam for students who want to try to place into Physics 207, which is not a BSE requirement, but which might be of interest to some students who really, really like physics. Most BSE students will take the 103-104 test if they want to try to place out of the BSE physics requirement.

Chemistry: If you received a 5 on the Chemistry AP exam, a 7 on the International Baccalaureate Higher Level exam, or an A or A* on the British A-levels, you will fulfill the BSE general chemistry requirement (CHM201/207) without taking a placement test. Make sure the score has been reported to Princeton!

Why should you take the chemistry placement test? A few possibilities:

1. If you studied chemistry at a high level but didn't take an AP exam, or if you took the AP exam and got a 4 or equivalent (e.g. IB-H 6, A-level B), but you think you know more than that score shows; if you took the SAT2 subject test in chemistry and got a very high score (like 750 or above), consider taking the placement test.
2. If you think you know a lot more chemistry and want to try to place out of general chemistry (which is a two-semester course sequence: CHM201/207 **and** CHM202) in order to jump into CHM 301 (Orgo) or CBE 245, then you need to take the departmental placement test.
3. If you have any other preparation in chemistry other than that mentioned above, or if you scored less than a 5 on the AP exam and still think you know more chemistry than the score shows, then you need to take the placement test, or otherwise take CHM 207 or 201.

Please note that per Chemistry departmental policy, taking the placement test is a **once-in-a-lifetime** opportunity. You may not take it a second time. Ever. If you have a 5 on the Chem AP exam but take the test to try for placement out of both CHM201/207 and CHM202 and fall short, we will still honor the 5 for the BSE requirement.

If you have taken any other exams (e.g. national school-leaving exams), or if you have taken college-level courses in physics or chemistry, but don't have an eligible AP/IB/A-level score, you must also take the departmental placement exam to fulfill B.S.E. requirements.

Information on the departmental placement tests in physics and chemistry is posted on the Path to Princeton website: <https://path.princeton.edu/content/stem-placement-testing-details>.

Math: In math, the situation is a bit different, since all B.S.E. students will continue to study math, and we can use your scores to make tentative placements which can be confirmed by the Math Placement Workshops on August 25. There is no placement test in Math. See the information below about Math placement.

Math accepts a score of 5 on the BC Calculus exam as fulfilling MAT103, which is an automatic Math 104/EGR 152 placement, but you might have gone further in calculus and might be a candidate for Math 201. See discussion below.

Chemistry

The one-semester B.S.E. general chemistry requirement may be satisfied with either an eligible test score or by taking CHM 207, 201, or MOL 214. How you use it depends on whether more chemistry will be required by your program of study.

If you have not placed out of the chemistry requirement: Plan on taking either Chemistry 201 or 207 in the fall of your first year. Note: CHM 201 and 207 are only offered in the fall. We consider 207 to be an excellent course especially for students who want to learn about applications of chemistry to engineering sciences. Students who are committed to majoring in Computer Science or ORFE and wish to get an early start by taking COS 126 in their first term are welcome to discuss the wisdom of deferring general chemistry to the fall of sophomore year with their advisers and with Dean Swanson. This is not a good idea in other engineering majors. We will also accept Molecular Biology 214 in fulfillment of this requirement. Please discuss with your adviser whether this is a viable option for your program of study. MOL 214 is challenging, and it is not an easier alternative to chemistry.

If you have fulfilled CHM201/207 through an eligible test score: If you do not plan on taking further chemistry (i.e. you are not interested in CBE, materials science, or premed), then it is acceptable to satisfy the requirement with an eligible test score and take no more chemistry courses. If you fulfilled one semester of general chemistry (CHM201/207) through an eligible test score and need to take more chemistry for your major, program, or premed requirements, then you must take CHM 215 so you can take Organic Chemistry in the fall of sophomore year. CHM 215 will be offered in spring 2026. Consider taking COS 126 or ECE 115 in the spot in the fall freed by not needing to take chemistry.

If you have fulfilled both semesters of general chemistry (CHM201/207 and CHM202) by taking the departmental placement test: If you do not plan on taking further chemistry (i.e. you are not interested in CBE, materials science, or premed), then you have satisfied the BSE chemistry requirement and can consider taking COS126 or ECE115 in the spot in the fall freed by not needing to take chemistry. If you are interested in Chemical and Biological Engineering, the strong recommendation of the CBE department is that take CBE 245 in the fall of your first year. Doing so will accelerate your entry into the CBE department and also show you whether this is the major for you sooner rather than later. You can then take Organic Chemistry in sophomore year.

Physics

The two-semester B.S.E. general physics requirement may be satisfied by taking the departmental requirement fulfillment test for Physics 103-104 or the placement test for Physics 207, or otherwise by taking PHY 103-104, PHY 105-106, or EGR 151-153. **You cannot place out of Physics 103-104 without taking the departmental requirement fulfillment test.** Note that you cannot place out of a single semester of Physics (e.g., 103) through the department requirement fulfillment test; you either place out of both semesters (PHY103-104) through the department tests or you must take PHY 103-104, PHY 105-106, or EGR 151-153.

If you did not take high school physics: Please consult with Dean Swanson about possible pathways that may require summer study before sophomore year.

If you do not fulfill PHY103-104 through the department requirement fulfillment test: Plan on taking Physics 103-104 or EGR 151-153. Most B.S.E. students will be taking these courses. If you especially like physics, have strong preparation, and would like a more theoretical treatment of the material, consider taking PHY 105. EGR 151 normally must be paired with EGR 152, Calculus II. If you are taking Math 201, you need to take Physics 103.

If you fulfill PHY103-104 through the departmental requirement fulfillment test: You have a choice. About half of the small number of B.S.E. students who satisfy the 103-104 requirement by the test elect to replace their physics courses with other subjects, either sophomore-level courses in a department that is a potential major or humanities/social science/language courses. The other half elect to take first-year physics to get a stronger physics foundation for their engineering-science courses.

Mathematics

The B.S.E. general math requirement consists of two semesters of single-variable calculus (Math 103-104), multivariable calculus (Math 201, 203, or 216/218), and linear algebra (Math 202, 204, or 217). Many B.S.E. students use placement to satisfy one or both of the single-variable calculus requirements. B.S.E. students may NOT take Math 175.

It is a really good idea to visit the Math department's [webpage on placement](#) and follow links to the various courses, then go to the "sample problems" and look at exams. If you think you should be placed in Math 104, take a look at the 103 sample exams. Can you solve most of the problems from start to finish (not just conceptualize how to solve them) **without a calculator**? Same for 201 versus 104. Don't look at the answers before attempting the problems.

Students who are taking Math 103, 104, or 201(203) this fall will first participate in the Math Placement Workshops on August 25. When you enroll in courses on August 26, you will be able to make an informed decision about which course is appropriate for you. You can make further adjustments during the first week of class.

If you have no calculus preparation: Please consult with Dean Swanson about possible pathways in math that may require summer study before sophomore year.

If you have no prior advanced coursework (e.g. AP/IB/A-levels/college course) in math: Start in 103 and move through the sequence in order: 103-104-201-202. This is not a problem, although in CBE it may be advisable to take differential equations in the summer after sophomore year.

If you have placement out of Math 103 (as indicated by AP 5 on BC calc, IBH7, A-level A): Start in 104 and proceed in order: 104-201-202. EGR 152 is an alternative to Math 104. If you got a 4 on BC calc or a 5 on AB calculus, then Math 104 or EGR 152 might be a reasonable place to consider starting, in our view (but you should also discuss with the math department during the Math Placement Workshops). Successful completion of Math 104 or EGR 152 will allow us to consider the BSE first-term math requirement fulfilled by exemption.

If you have taken a full year or more of single-variable calculus, equivalent to Math 103-104: A good indication of this would be a solid 5 on the BC calculus, an A* on British A-levels, or some other indicator of full mastery of single-variable calculus. A lot depends on how confident you are in your mastery of single variable calculus at the level of the end of [Math 104](#). If you have studied [Taylor series](#), then you are probably at or close to this level. The most common place is to start in Math 201 and then take Math 202 in the spring to complete the general B.S.E. math requirement. If you feel a bit shaky in your basic calculus, or if there were some gaps in your topical coverage (e.g. Taylor series), you can choose to start in Math 104. No problem with that. The Math Placement Workshops will help you make this decision. Successful

completion of Math 201/203 will allow us to consider the BSE first- and second-term math requirements fulfilled by exemption.

If you have a really strong background in math and really, really like math: You might consider becoming a Math major, although we'd prefer that you stay in the B.S.E. program and explore math more deeply through ORFE or COS. The way to test this interest is to take Math 215 or Math 216. Although Math 215 does not fulfill a B.S.E. requirement, we will accommodate a certain amount of exploration in the fall of the first year. If you remain a B.S.E. student after taking 215, please understand that you still must complete courses in multivariable calculus and linear algebra. 216 is acceptable for the B.S.E. multivariable calc requirement. In addition to the standard sequences above, the following combinations are possible: 216-217, 217-218, even 217-201.

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If you have already taken multivariable calculus and/or linear algebra *at the college level* and want to try to use these courses to satisfy the B.S.E. requirements, you must contact Dr. Ana Menezes, the Math Placement Officer (see <https://www.math.princeton.edu/undergraduate/placement>). Dr. Menezes will determine whether you have indeed covered these topics adequately at the level expected at Princeton. If such a determination is made and **confirmed in writing**, then a notation will be made on your record that one or both of these requirements has been satisfied. Bringing examples of graded assignments and exams, along with your syllabus, from previous college-level courses in these subjects to show the Math Placement Officer will facilitate the conversation. Pre-college courses in differential equations may not be used to satisfy departmental requirements for MAE 305.

Computer Science

The B.S.E. computing requirement must be satisfied by taking a course at Princeton, normally COS 126. If you have substantial background in computer science already and would prefer to satisfy the requirement by taking COS 217 or 226, you should take the online placement exam. See details here: <https://path.princeton.edu/content/stem-placement-testing-details>. Placement into 217 or 226 does not in itself fulfill the B.S.E. computing requirement. You must take a course. Take note of ECE 115, which is an alternative to COS 126 to fulfill the B.S.E. requirement.

Humanities and Social Sciences

The B.S.E. humanities and social science requirements and the distribution areas covered by them are fulfilled by taking courses. AP test scores cannot be used to fulfill them. This includes the option of using foreign language study at the 107/108 level or higher as one of your four distribution areas. Unlike the A.B. language proficiency requirement, the B.S.E. distribution area in Foreign Language is fulfilled by taking a language course at the 107/108 level or higher. If you think you may want to take a language course in a language you've previously studied, do take a language placement test. See details here: <https://path.princeton.edu/student-documents/information-regarding-fulfillment-language-requirement>

Questions?

B.S.E. students should direct questions about placement to the Associate Dean for Undergraduate Affairs, Dean Cecily Swanson. To make an appointment, please contact Ms. Traci Miller by e-mail at BSEprogram@princeton.edu.